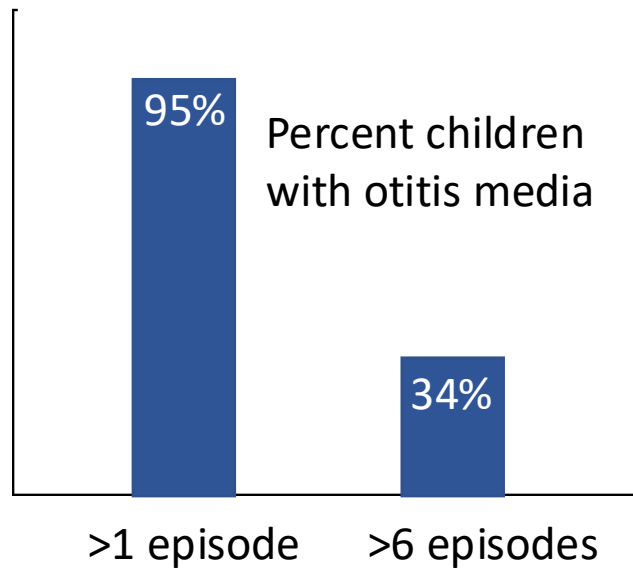


Reducing U.S. Children's Antibiotic Exposure via Non-Invasive Immune Therapy

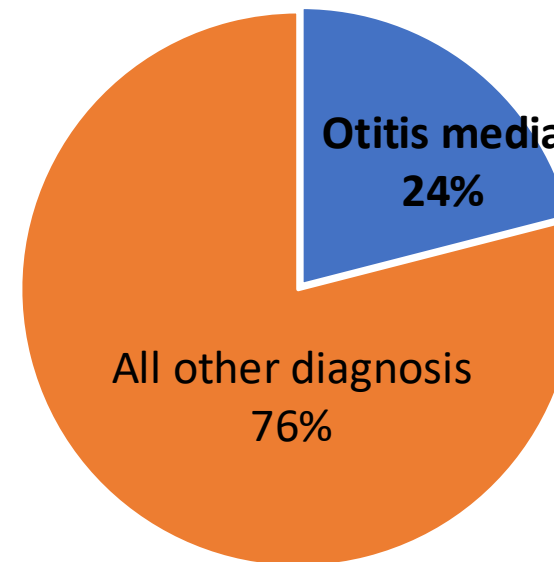
- Key Team: Drs. Rong Yang, Di Yang and Wenjing Tang
Yang Lab @Cornell (<https://theyanglab.com/>)
- Project Period: July. 1, 2025 - June. 30, 2026
- 6-Month mid-term review: Feb 23, 2026

Market Needs: otitis media (OM) is the #1 reason for pediatric antibiotic prescription



Faden et al., *Pediatr. Infect. Dis. J.*, 1998
<http://www.nfid.org/>
Teele et al., *J. Infect. Dis.*, 1989

Antibiotic prescription to U.S. children



Global mortality: **21,000** per year

Competition Addressed: systemic antibiotic treatment causes side effects and breeds resistance

Oral antibiotics multiple times per day for at least one week



Cox et al., Nat. Rev. Endocrinol. 2015
Risnes et al., Am. J. Epidemiol. 2011

Poor Compliance

Administration challenges for children

Side Effects

Rashes
Vomiting
Diarrhea

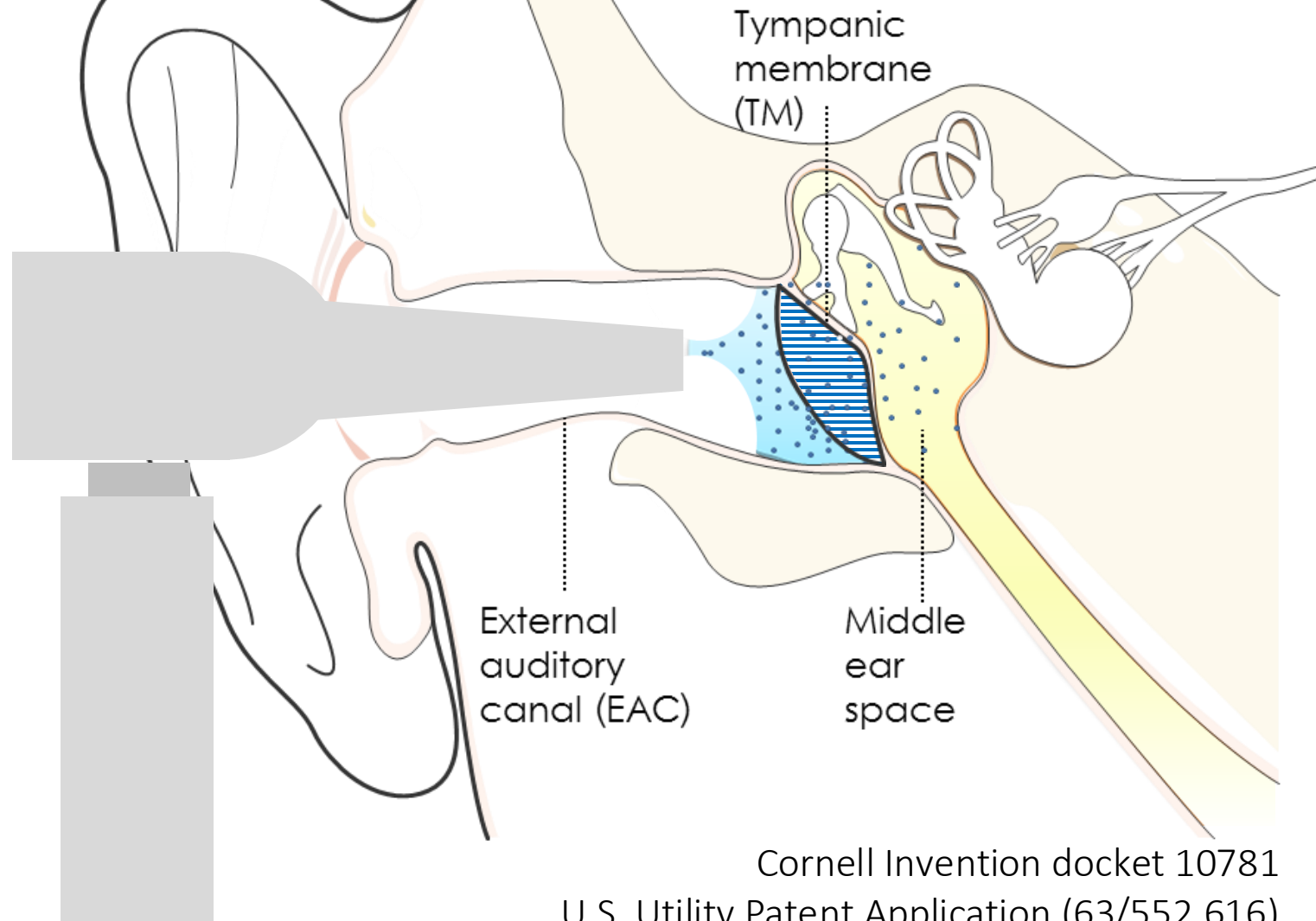
Alter Microbiomes

Asthma
Obesity

Antibiotic Resistance

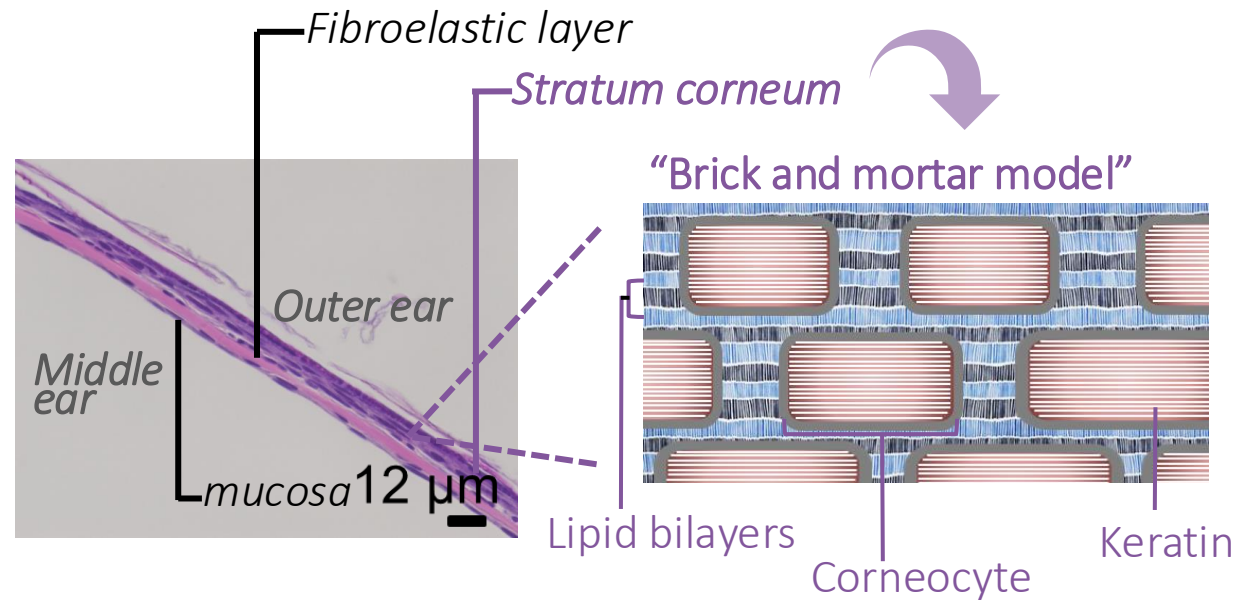


Technology: proprietary trans-tympanic drug delivery platform

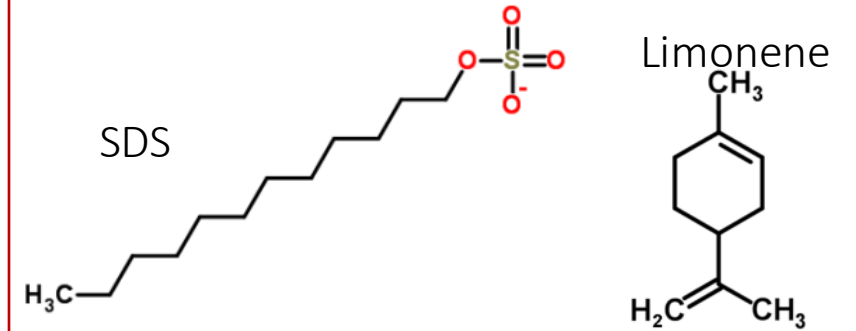


Cornell Invention docket 10781
U.S. Utility Patent Application (63/552,616)

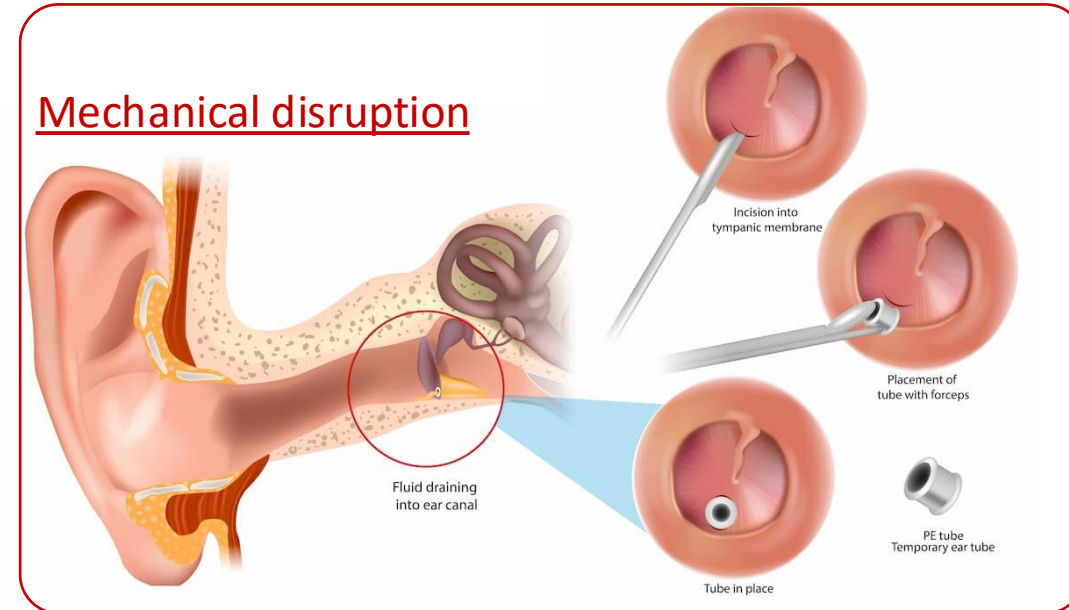
Why hasn't there been an eardrop for AOM?



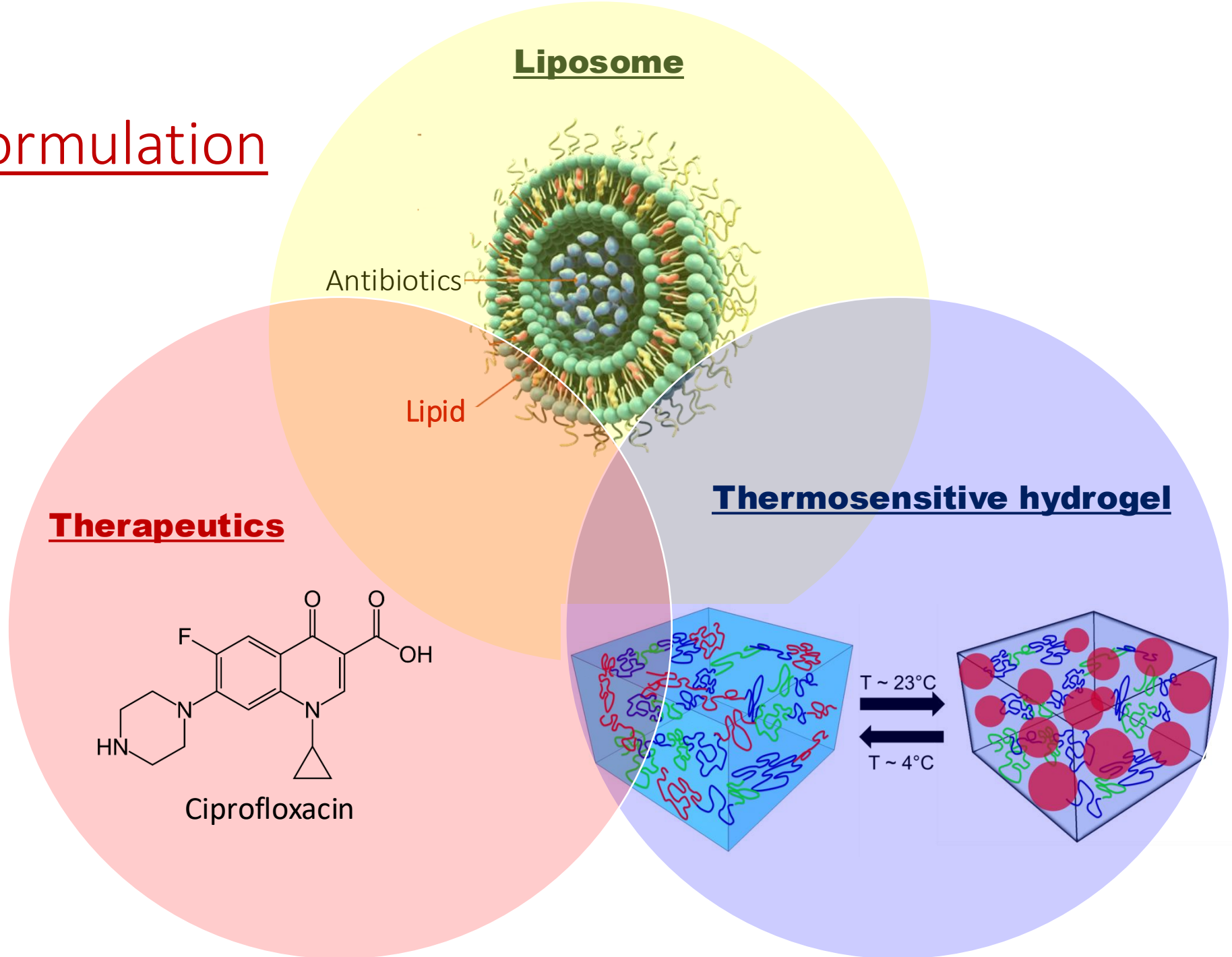
Physicochemical disruption



Mechanical disruption

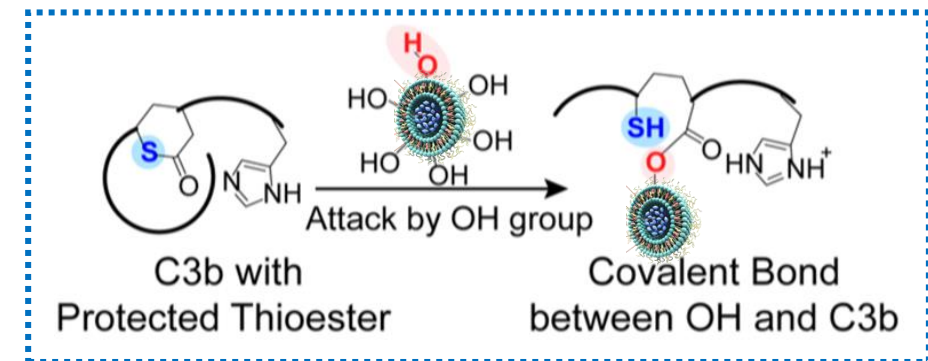
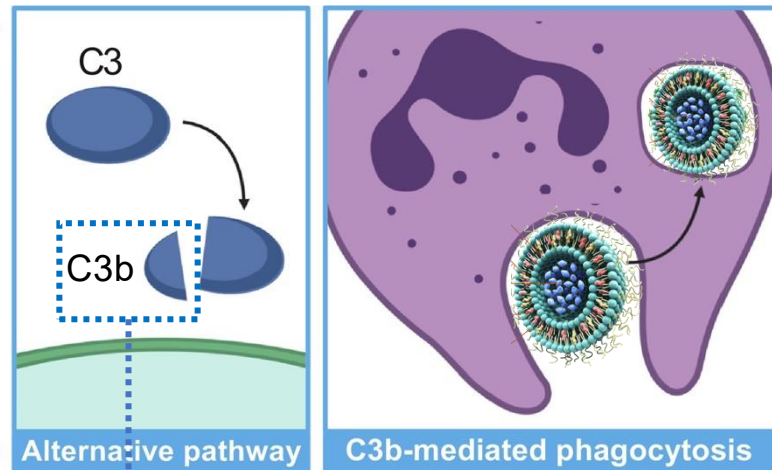
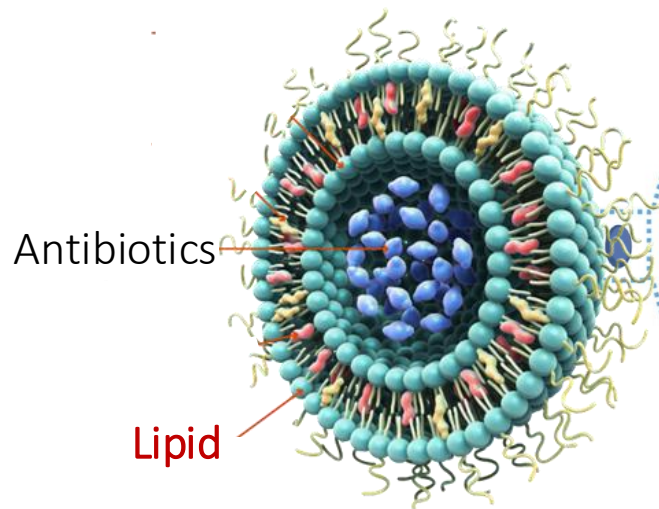
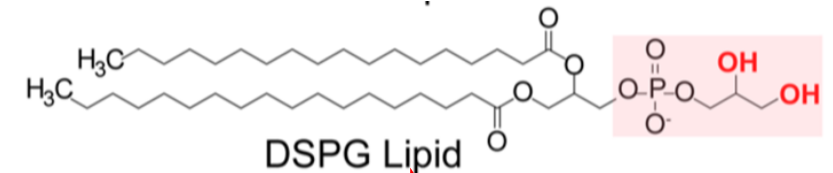
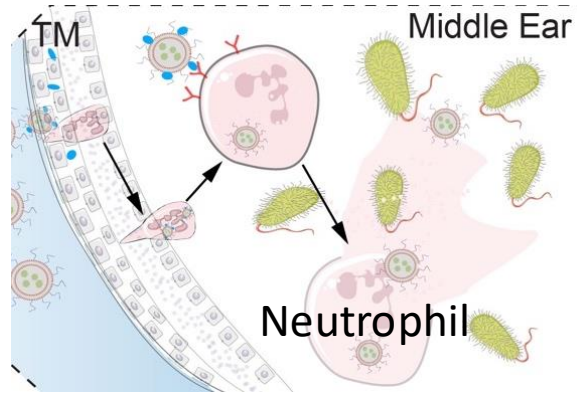
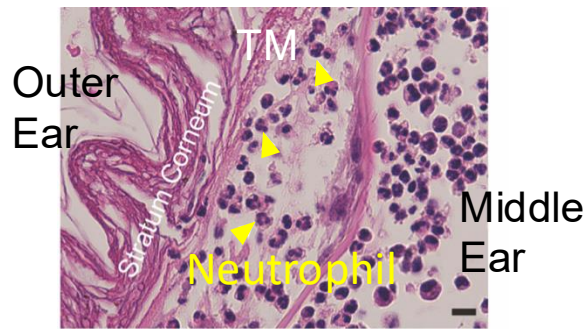


Our Formulation

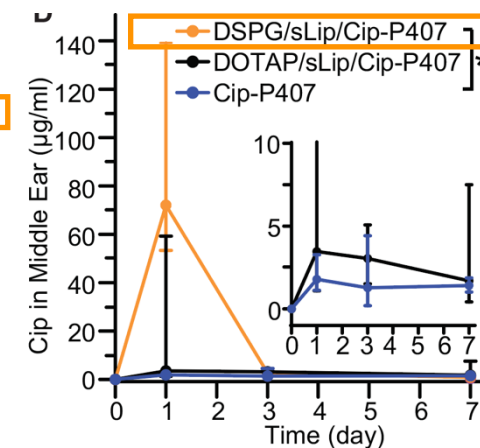
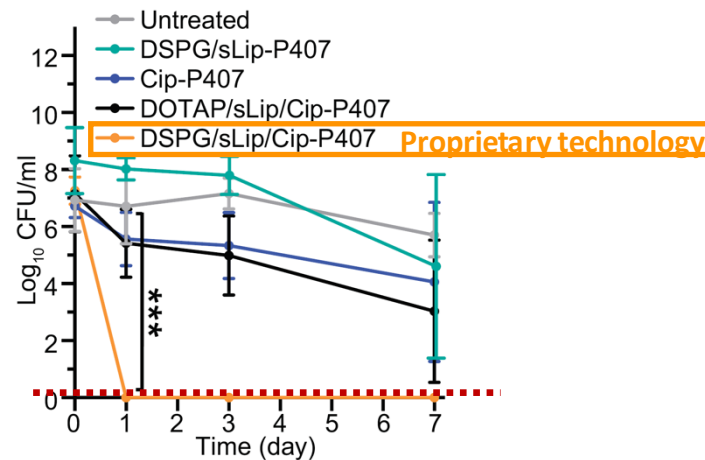
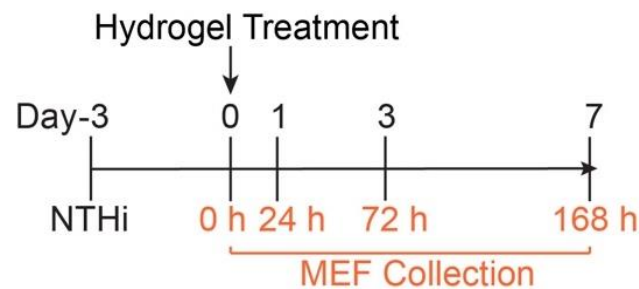
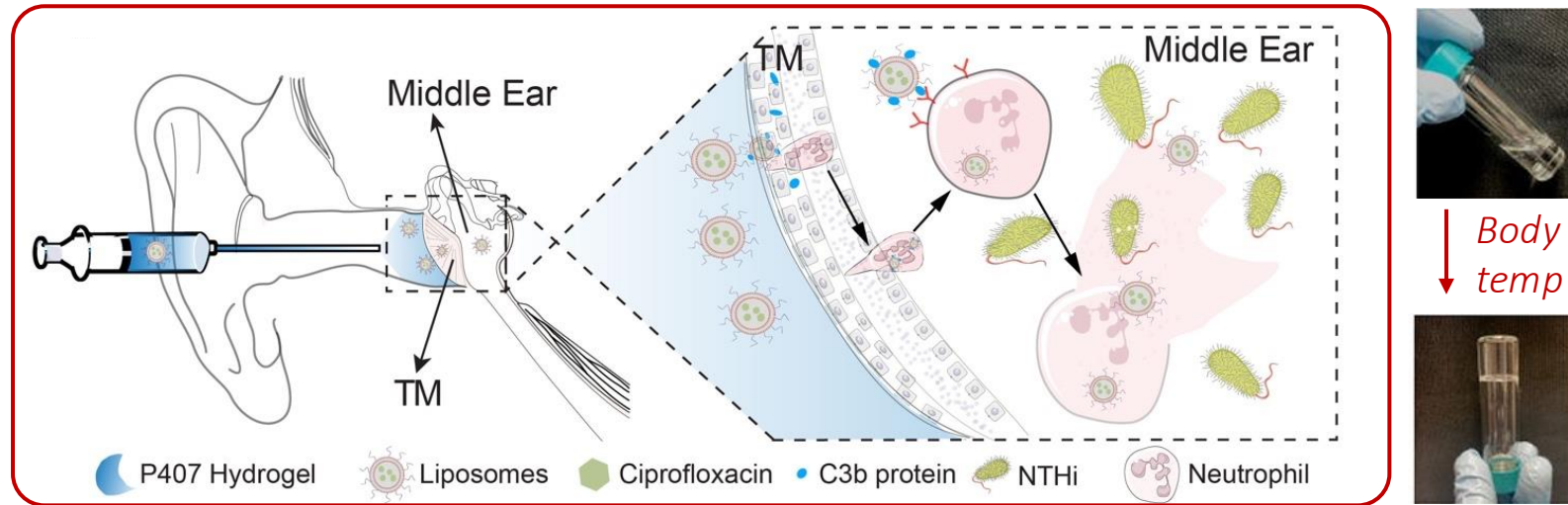


Our technology: Antibiotic-containing and neutrophil-hitchhiking liposomes for non-invasive treatment

Proposed approach: neutrophil-enabled delivery



Efficacy: single-dose, sustained delivery over 7 days via a thermoresponsive hydrogel



DSPG/sLip/Cip-P407 eradicated infection within 24 hours with 1 mg per kg body weight (625 times less than oral one) in an FDA-recognized preclinical animal model.

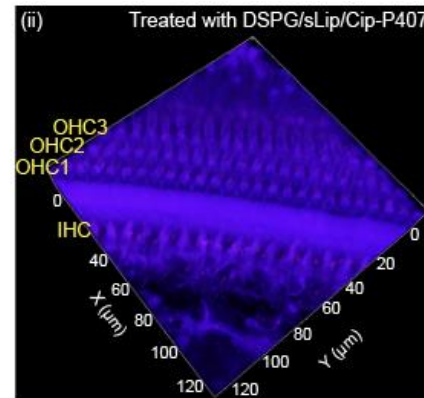
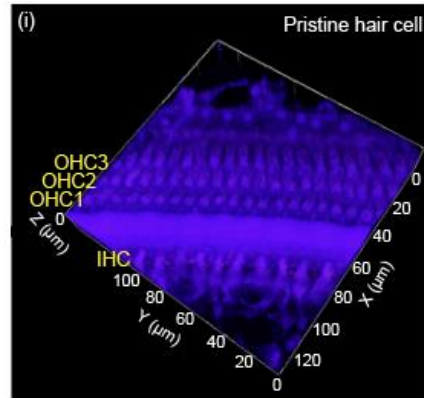
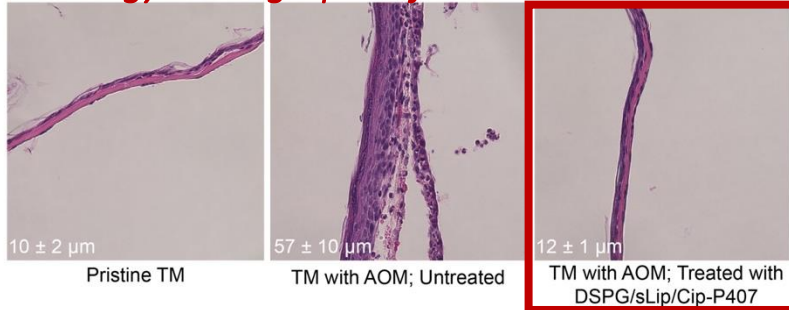
NTHi, MIC < 0.1-0.5 µg/ml

Technology: bypassing systemic antibiotic exposure to avoid side effects and hearing loss

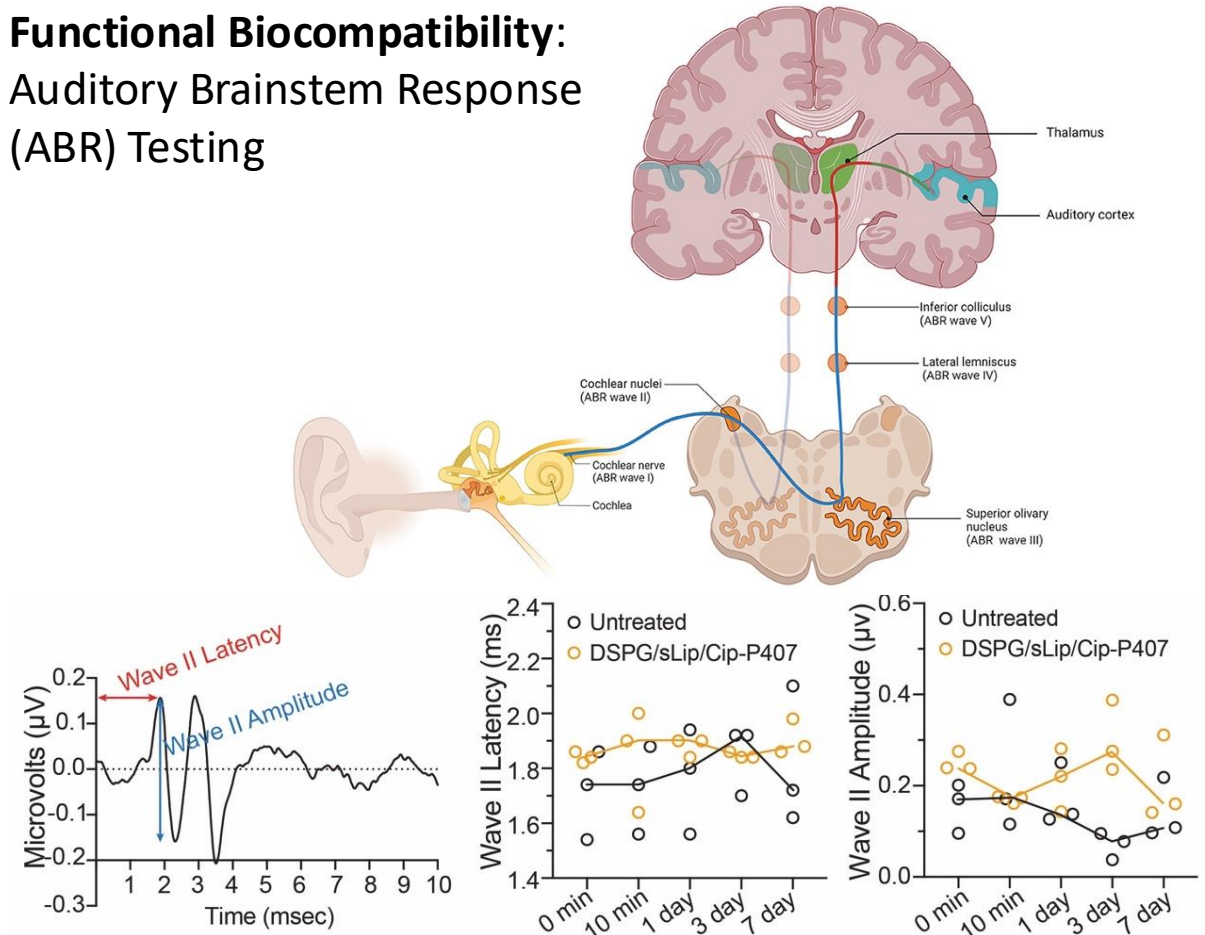
DSPG/sLip/Cip-P407 reversed TM inflammation after 7 days of treatment.

Tissue Biocompatibility:

Histology micrographs of the TM



Functional Biocompatibility: Auditory Brainstem Response (ABR) Testing

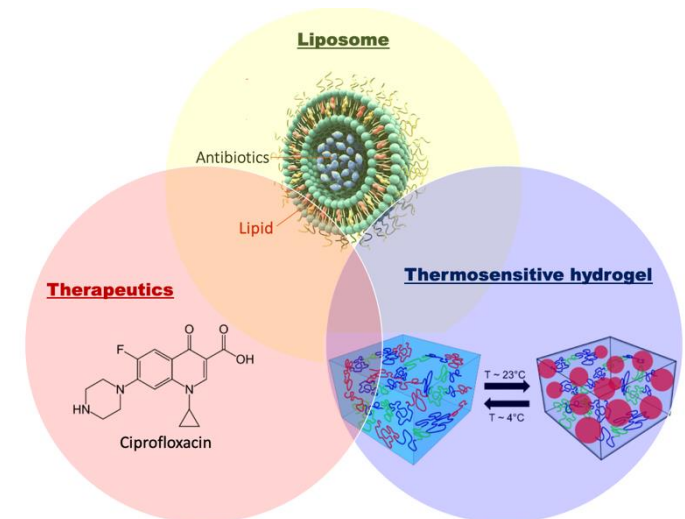


Proof of Concept

- ✓ Preliminary safety and efficacy data in FDA-accepted preclinical animal model

Value proposition

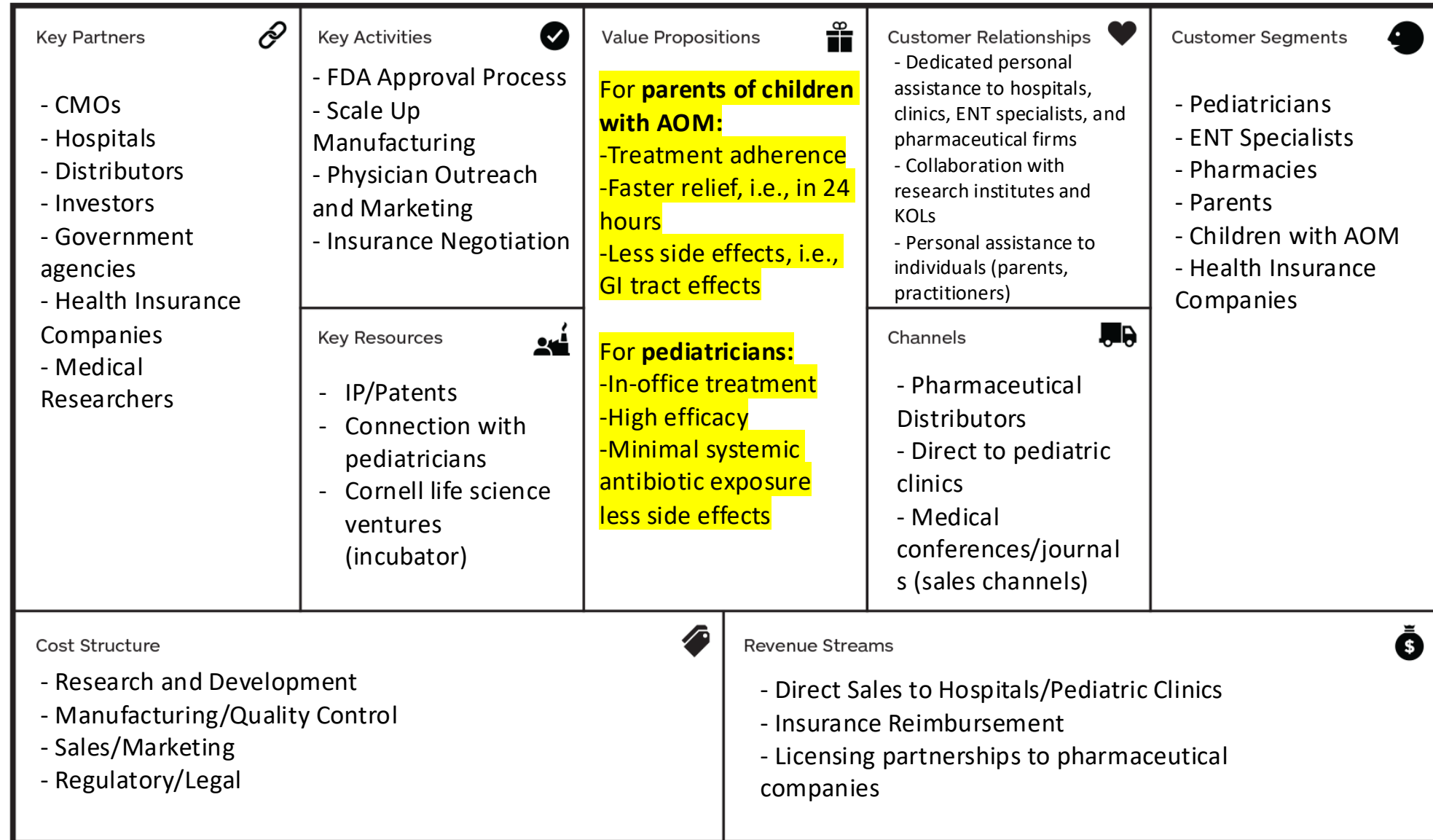
- ✓ Improve patient compliance
- ✓ Low antibiotic exposure
- ✓ Rapid cure: no hearing loss and mitigated developmental defects of speech and language
- ✓ No systemic side effects
- ✓ Compatible with virtually any antibiotics



Personal Trainings / Courses / Connections

- ABI course, Aug 26 - Dec 9, 2025 (led by Loren Busby)
- Regional I-Corps course, Oct 20 – Nov 14, 2025 (led by Emily Marino and Matthew Quimby)
- Dr. Madeleine Drusin, Associate professor of otolaryngology, meet monthly since Mar 5, 2026
- Where to Play workshop, Feb 23 – Mar 11, 2026 (led by Sharon Tal-Itzkovitch)

Business Model Canvas (Regional I-Corps)



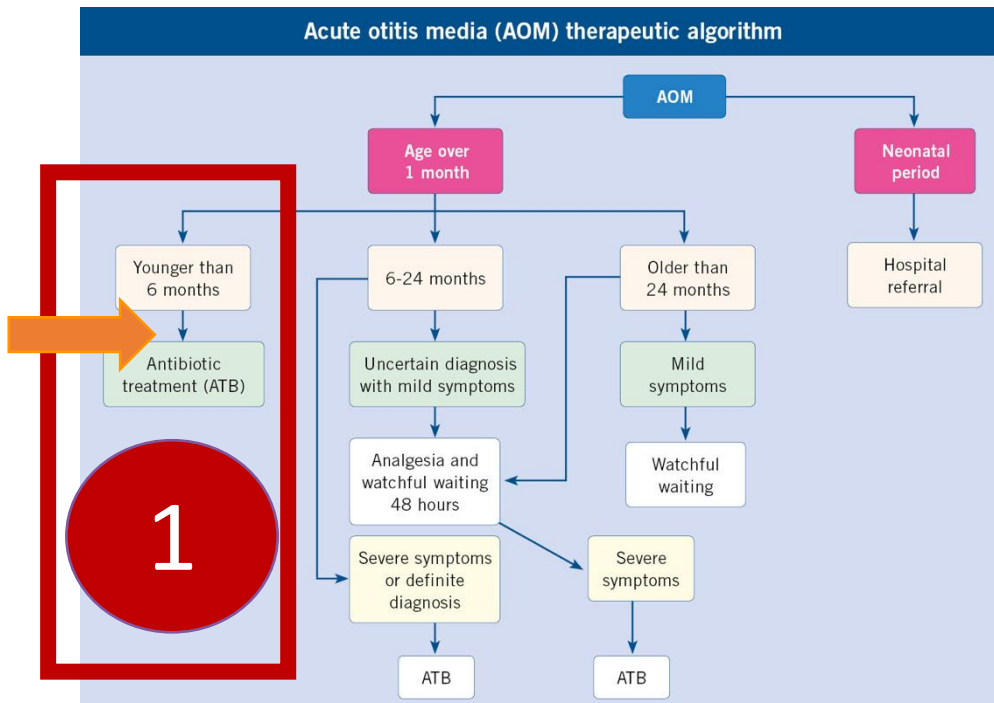
Proposed Activities to be Funded, Anticipated Milestones & Deliverables

	2025 Q3	2025 Q4	2026 Q1	2026 Q2
Aim 1: Technology development	-Establishing collaboration with Good Manufacturing Practice (GMP) and GLP-compliant facilities	-Validating the formulation manufacturing protocol and assessing stability under different storage conditions	-Establishing collaboration with physician-scientists for double-blinded preclinical experiments	-Performing detailed efficacy and hearing tests -Characterizing systemic drug exposure
Aim 2: Business plan development	-Assessing market size and competition -Customer discovery		-Physician interviews -Identifying regulatory and reimbursement strategy	
Aim 3: Company operation	-Company registration and identifying the founding team		-company formation -identifying operating space -identifying strategic partners -CTL license	
Aim 4: Fundraising	-Application to SBIR (NSF and NIH) -Bill and Melinda Gates Foundation: Innovations for Gram-Negative Antibiotic Discovery -CARB-X Funding (Combating Antibiotic-Resistant Bacteria)		-Startup competitions (e.g., HLTH USA 2025, MIT 100k) -Build connections	-Leverage connections for angel investors, venture capital firms, strategic investors/partners, crowdfunding

Proposed Activities to be Funded, Anticipated Milestones & Deliverables

	2025 Q3	2025 Q4	Achievements
Aim 1: Technology development	-Establishing collaboration with Good Manufacturing Practice (GMP) and GLP-compliant facilities	-Validating the formulation manufacturing protocol and assessing stability under different storage conditions	- Collaborating with Evonik (CMC partly) for the GMP for the manufacturing of the formulation - Incubator space (in progress) for formulation test - CRO collaboration (animal model, focus on biosafety for pre-IND) - regulatory consultancy, pathway exploration (510(k) and 505(b) discussion)
Aim 2: Business plan development	-Assessing market size and competition -Customer discovery		- shown in the following slide
Aim 3: Company operation	-Company registration and identifying the founding team		    Di Yang, PhD Prof. Rong Yang, PhD Wenjing Tang, PhD CEO & CO-FOUNDER CSO & CO-FOUNDER CHIEF TECH OFFICER
Aim 4: Fundraising	-Application to SBIR (NSF and NIH) -Bill and Melinda Gates Foundation: Innovations for Gram-Negative Antibiotic Discovery -CARB-X Funding (Combating Antibiotic-Resistant Bacteria)		- Submitted NIH SBIR Phase I in Sep 2025, plan to resubmit in Apr 2026 - Submitted NSF SBIR Phase I in Dec 2025 - Submitted CAT Grant in Nov 2025 - Submitted Resnick Prize in Mar 2026

Value proposition and patient profile based on pediatrician interviews



i Brú, J. de la Flor. *Pediatría Integral* 352 (2022): 353-368.

Target the first-line treatment

#1 challenge among infants (<6 months): patient compliance

- ✓ 1st line treatment for Infants
 - ✓ Improve patient compliance
 - ✓ Reduce antibiotic exposure

Market assessment: targeting infants less than 6 months in the U.S. (first line treatment, direct Abx intervention)



AOM infants (< 6 months): 2 Million

Market assessment: \$400/dose

Short-term market penetration target: 5%

Peak U.S. revenue potential: 40 Million

Peak-sale multiple: 3x

Preclinical stage risk adjustment: 10%

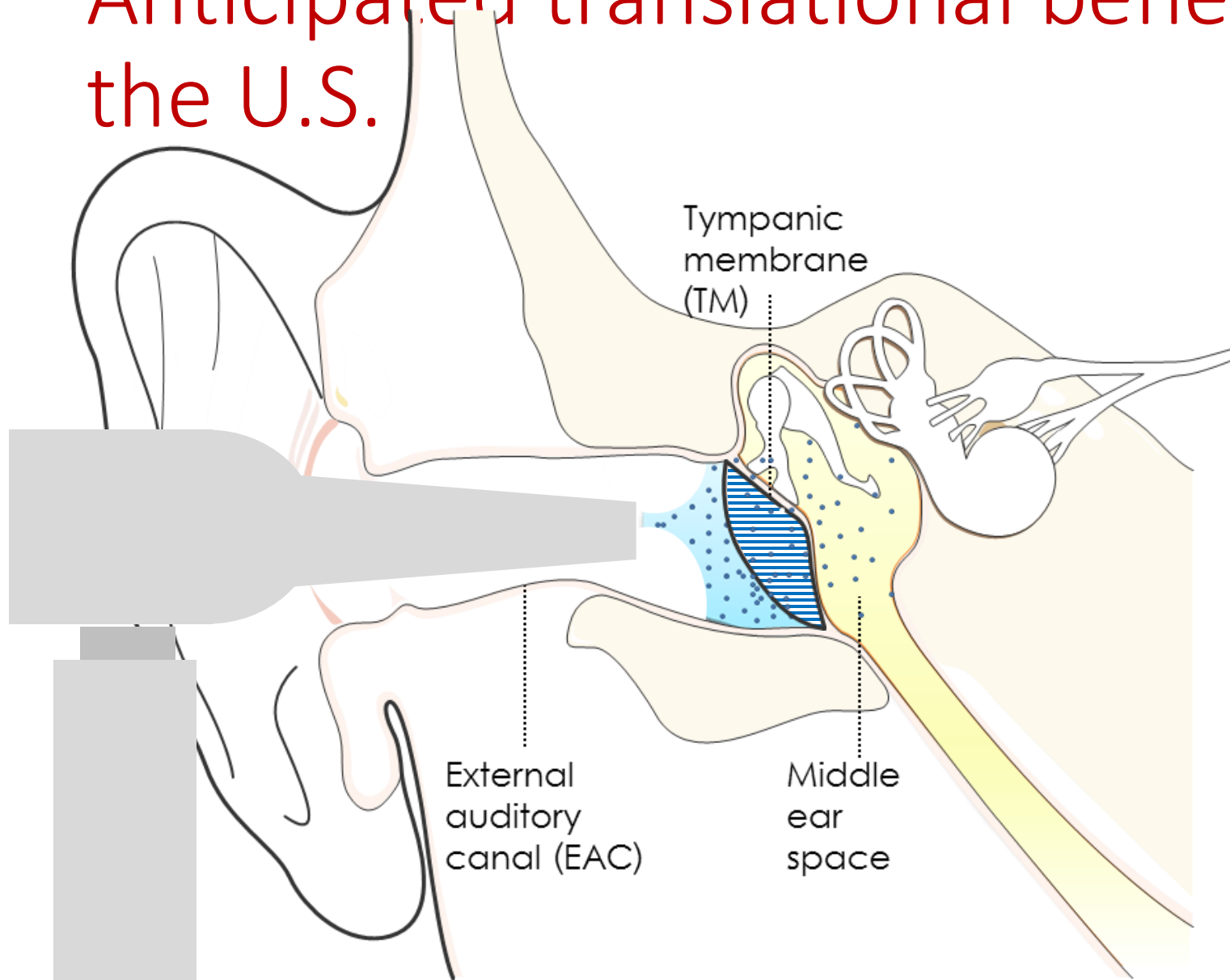
Current risk-adjusted value: 12 Million

Why We are Great and Our Competition sucks

COMPETITION-- (Oral antibiotics)

	We can, it can't	It can, we can't
(Competitor 1) Oral amoxicillin	- Local, high drug concentration directly at infection site (middle ear) - Bypass systemic circulation → reduced systemic side effects - Effective against resistant/localized biofilms - Non-invasive, targeted transtympanic delivery	- Systemic treatment (can address concurrent infections elsewhere) - Easy administration (oral dosing, no device needed) - Widely accepted standard of care in U.S. - Lower upfront procedural/technology barrier
(Competitor 2) Other oral antibiotics (e.g., augmentin)	- Avoid GI-related side effects (e.g., diarrhea, microbiome disruption) - Reduce antibiotic overuse → antimicrobial stewardship benefit - No compliance issue with multi-day oral dosing	- Broad-spectrum systemic coverage - Familiar prescribing patterns for clinicians - No need for specialized delivery system - Insurance reimbursement already established

Anticipated translational benefit to children in the U.S.



Patient compliance: less pediatric hospitalization and Abx resistance

~1,000-fold reduction in antibiotic usage in OM treatment

Avoiding systemic side effects on GI, nervous system, and hearing

Rapid cure: less hearing loss or developmental defects of speech and language

✓ Potential to treat respiratory infections

**: Confidential disclosure filed on 6/7/2023*

Acknowledgment

OTO Biotech Inc:

Core Team:

Prof. Rong Yang, CSO

Dr. Di Yang, CEO

CSLV Team:

Center Staff:

Ying Yang, Interim Director

Yosey Hagai, Scientific Director

Weiwei Yan, Lab Operations Manager

Institute of Biotechnology Liaison:

Kurt Deriziotis

Regulatory Advisor:

Gregory Frykman, MD

Special Thanks To:

Lou Walcer, Former Director

CTL Team:

Leadership:

Alice Li, Executive Director

Licensing & Business Development:

Stephen Nowak, Interim Manager

Technology and Venture Initiatives & Engagement:

Lynda Inséqué, Director

Aaron Delahanty, Manager

WCM:

BioVenture eLab:

Loren Busby, Director

Pediatric Otolaryngology:

Madeleine Drusin, M.D., M.Sc.

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VENTURES

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Weill Cornell Medicine